

CSE220 Resources [MUNR]

	A	B	C	D	E	F	G	H
1	CSE220: Data Structures (Spring 2026)							
2								
3								
4			Course Outline: https://docs.google.com/document/d/1MoFJiBQLEAPtRSiXDvFR7nA9PKIhvTlxwHwtbrGNXOw/edit?tab=t.0					
5			Course Lecture Plan (Central): https://docs.google.com/spreadsheets/d/1GRfHLKw6z0bTJV8VSiEGM1f1vgjdDPAdbalsMlm8F2o/edit?gid=0#gid=0					
6			Course Textbooks: https://drive.google.com/drive/folders/1a4J0GkumNjzPgWdBzKHpk2t1aA--S3s?usp=drive_link					
7			Previous Semester Questions: https://drive.google.com/drive/folders/1N4r56k2m4pouv3y9GSUt-oCZhmgcJVfm?usp=sharing					
8			Course Materials Drive Link: https://drive.google.com/drive/folders/1_qWM8NTdkxIHhuUCH-wx9l8rMAP73lxb?usp=sharing					
9			Class Recording Playlist: https://www.youtube.com/playlist?list=PLYv5ovgT8QC79CTFkvdHo7ZTcM0ZJ3ALA					
10			Faculty Details: https://cse.sds.bracu.ac.bd/faculty_profile/212/abm_muntasir_rahman					
11								
12								
13	Lecture No.	Topic	Sub Topics (covered in class)	Lecture Notes	Self Study	Supplementary Lectures	Code Snippets	Practice Sheet
14	0	Ice Breaking	-	-	-	-	-	
15	1	Introduction to Data Structures & Time Complexity	1. Data structures and it's purpose 2. Complexity Primer	■ Chapter_0.pdf	1. Topics covered in class	■ Complexity Primer (09/02/26)	-	■ Time and Space Complexity Examples
16	2	Linear Array Operations	1. Purpose of Linear Arrays 2. How Arrays are stored in memory 3. Accessing 4. Iteration 5. Resize 6. Copy 7. Shift (Left & Right) 8. Rotate (Left & Right) 9. Size 10. Insertion 11. Deletion 12. Linear vs Binary Search	■ Chapter_1.pdf	1. Implementing code of all the operations mentioned in Sub Topics section along with step by step understanding 2. Array Reversing (In-place, out-place) implementation with step by step understanding 3. Solve tasks from practice sheet/previous questions/labs/online judge etc. 4. Bubble Sort/Selection Sort/Insertion Sort	■ 1.2 Array Operations - Traversal, Insertion Explanation with C Program DSA Course ■ 1.3 Array Operations Deletion from Array Explanation with Code Data Structure	■ 1. Linear Array	■ Array Practice Problems
17	3	Introduction to Multidimensional Arrays & 2D Array Operations	1. Purpose of Multidimensional Arrays 2. How multidimensional arrays are stored in memory 3. Mapping dimension of linear index to 2D/3D/4D index and vice-versa 4. Accessing 5. Iteration (Row & Column) 6. Summation (All, Row & Column) 7. Swapping (Row & Column) 8. Addition (Primary and Secondary Diagonal, Two matrices of same dimension) 9. Previous semester's problem solving		1. Linear Array index to Multidimensional Array index translation math (example can be found in book) 2. Implementing code of all the operations mentioned in Sub Topics section along with step by step understanding 3. 2D Matrix Multiplication implementation with step by step understanding 4. Solve tasks from practice sheet/previous questions/labs/online judge etc.	https://youtu.be/Q3aANguLjyw ■ 2D Array Operations (Addition & Multiplication) & Problem Solving (17/02/26) - YouTube	■ 2. 2D Array	
18	4	Introduction to Singly Linked Lists & Singly Linked List Operations	1. Advantages and disadvantages of Arrays & Linked Lists 2. Creating a list from an array using Node class 3. Iteration 4. Counting elements in list 5. Retrieving element from an index (elemAt) 6. Retrieving node from an index (nodeAt) 7. Update the value of an index (updateValue)	■ Chapter_2.pdf	1. Implementing code of all the operations mentioned in Sub Topics section along with step by step understanding 2. Linked List Rotation (Left & Right) implementation with step by step understanding 3. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ 2.5 Insertion of a Node in Linked List(at Beginning,End,Specified Position)with Code DSA Tutorials ■ 2.6 Deletion of a node from Linked List (from beginning, end, specified position) DSA Tutorials ■ 2.8 Reverse a Linked List - Iterative Method Data Structure Tutorials ■ Singly Linked List Operations (02/07/25)	■ 3. Singly Linked List	■ Singly Linked List Practice Problems
19	5	Singly Linked List Operations	1. Searching 2. Insertion 3. Deletion 4. Reversing 5. Previous semester's problem solving					
20	6	Linked List Variations and Introduction to DHDCL	1. Types of Linked List 2. Creating a DHDCL from from an array using DoublyNode class 3. Iteration 4. Counting elements in list		1. Implementing code of all the operations mentioned in Sub Topics section along with step by step understanding 2. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ Variations of Linked List (30/06/24)	■ 4. DHDCL	■ Doubly Linked List Practice Problems
21	7	DHDCL Operations	1. nodeAt 2. Insertion 3. Deletion					
22	8	Stack	1. Introduction to Stack 2. Stack Operations (Push, Pop & Peek) 3. Stack Applications 4. Stack Implementation	■ Chapter_3.pdf	1. Implementation of Stack using Linked List 2. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ Stack (02/03/26) ■ Stack Problem Solving (02/03/26)	■ 5. Stack	■ Stack & Queue Practice Problems
23	9	Queue	1. Introduction to Queue 2. Queue Operations (Enqueue, Dequeue & Peek) 3. Queue Applications 4. Queue Implementation		1. Implementation of Queue using Linked List 2. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ Queue	■ 6. Queue	
24	10	Hashing	1. Introduction to Hashing and Hashtable 2. Collision Handle (Forward Chaining) 3. Hashing Example 4. Hashing Implementation	■ Chapter_4.pdf	1. Implementation of Hash Function & Inserting in a Hash Table 2. Implementation of Delete Function 3. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ Hashing (05/12/2024)	■ 7. Hashing	■ Hashing Practice Problems
25		Quiz 1	Solution Link: ■ [CSE220-11] S26 Quiz 1 Solve.pdf [GSuite Sign-In Required]					
26		Quiz 2	Solution Link: ■ [CSE220-11] S26 Quiz 2 Solve.pdf [GSuite Sign-In Required]					
27	11	Recursion	1. Introduction to Recursive Programming 2. Recursive function application: Factorial & Fibonacci 3. Recursive function application: Fibonacci with memoization, Exponent, Binary Search, Array Printing: Forward & Backward, Sum of Array	■ Appendix_A_Recursion.pdf	1. Topics covered in class 2. Linked List operations using Recursion approach 3. Implementation of practice problems	■ Recursion (13/04/25)	■ 8. Recursion	■ Recursion Practice Problems
28	12	Tree	1. Introduction to Tree 2. Tree Applications 3. Terminologies of a Tree 4. Characteristics of a Tree	■ Chapter_5.pdf	1. Understanding tree terminologies and characteristics	■ Tree & Binary Tree Introduction (06/12/25)	-	-
29	13	Binary Tree (BT)	1. Introduction to Binary Tree 2. Binary Tree Traversal (Preorder, Inorder, Postorder) 3. Types of Binary Tree 4. Binary Tree Implementation: Creation/Traversal/Height/Depth/Count /Mirror/Problem Solving etc.		1. Implementation of binary tree creation, traversal, depth, level, types of binary tree using Recursion 2. Implementation of binary tree practice problems 3. Solve tasks from practice sheet/previous questions/labs/online judge etc.	■ Binary Tree (15/09/24) ■ Binary Tree Implementation & Problem Solving (14/12/2024) ■ Binary Tree Problem Solving (19/08/25)	■ 9. Binary Tree	■ Binary Tree Practice Problems
30	14	Binary Search Tree (BST)	1. Introduction to BST 2. BST Operations: Creation/Insertion, Removal 3. BST Balancing 4. BST Implementation	■ Chapter_6.pdf	1. Step by step simulation and understanding of BST Operations: Creation/Insertion, Deletion, Balancing, Searching 2. Implementation of all BST Operations	■ Binary Search Tree (17/09/2024)	■ 10. Binary Search Tree	■ Binary Search Tree Practice Problems
31	15	Heap	1. Introduction to Heap 2. Heap Operations: Creation/Insertion, Deletion 3. Heap Sort 4. Heap Implementation		1. Step by step simulation and understanding of Heap Operations using both Max and Min Heap 2. Step by step simulation and understanding of Heap Sort 3. Implementation of all Heap Operations	■ Heap (31/12/24)	■ 11. Heap	■ Heap Practice Problems
32	16	Graph	1. Introduction to Graph 2. Graph Applications 3. Graph Terminologies & Types 4. Graph Representation: Matrix & List 5. Graph Implementation	■ Chapter_7.pdf	1. Graph Implementation: Adjacency List & Matrix 2. Understanding Graph Terminologies	■ Graph Basics & Representation (29/12/2024)	■ 12. Graph	■ Graph Practice Problems
33	17	Graph Traversal	1. Breadth First Search (BFS) 2. Depth First Search (DFS)		1. Step by step simulation and understanding of Graph Traversal Algorithms 2. Implementation of Graph Traversal Algorithms	■ Graph Traversal (07/05/2025)	■ 13. Graph Traversal	
34		Quiz 3	Solution Link: ■ [CSE220-11] Quiz 3 Solve.pdf [GSuite Sign-In Required]					
35		Quiz 4	Solution Link:					
36		Quiz 5	Solution Link:					